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語意通訊中多模態融合與訊號偵測之研究

Is Detection Necessary? Rethinking Multimodal Fusion in Task-Oriented Semantic Communications

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本論文係 劉宗翰 （學號 R12944009）在國立臺灣大學資訊網路與多媒體研究所完成之碩士學位論文，於民國 114 年 9 月 26 日承下列考試委員審查通過及口試及格，特此證明。

The undersigned, appointed by the Graduate Institute of Networking and Multimedia on 26 September 2025 have examined a Master's Thesis entitled above presented by TSUNG-HAN LIU (student ID: R12944009) candidate and hereby certify that it is worthy of acceptance.

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Author *Tsung-Han Liu*

摘要

隨著人工智慧驅動的應用與邊緣裝置迅速普及，無線通訊系統正面臨沉重的傳輸與計算負擔。任務導向的語意通訊（Semantic Communication, SemCom）因能在多樣化資料型態下，同時利用語意層級資訊與任務層級效能而逐漸受到關注。然而，現有多數多使用者方法仰賴明確的多使用者訊號偵測，這不僅增加系統複雜度與延遲，亦會在不利的通道條件下導致推論效能下降。我們的初步實驗顯示，對於多模態任務導向的語意通訊而言，訊號偵測並非不可或缺，原因在於語意資訊本身具有內在的相依性與互補性。基於此觀察，我們提出一個名為 DIRECT 的多模態語意通訊框架，在傳輸過程中實現多使用者資訊融合，從而消除接收端對明確訊號偵測的需求。為進一步提升任務效能並降低來自貢獻不均之冗餘特徵所造成的干擾，我們設計了一種語意加權的重要特徵傳輸機制（Semantic-Weighted Importance Feature Transmission, SWIFT）。透過 SHAP 為基礎的分析，SWIFT 能選擇性地傳輸與任務最相關的高價值特徵，有效減少語意冗餘並提升推論準確率。實驗結果顯示，DIRECT 在影片情感分析任務上的準確率，相較於不含訊號偵測的傳統語意通訊方法可提升約 2%–5%，相較於採用訊號偵測的方案則可提升約 10%。

關鍵字：

語意通訊、任務導向、多用戶通訊、多模態資料、訊號偵測、夏普利值

Abstract

The rapid proliferation of AI-driven applications and edge devices has imposed heavy transmission and computation demands on wireless communication systems. Task-oriented semantic communication has emerged to jointly leverage semantic-level information and task-level performance across diverse data types. However, most existing multi-user approaches rely on explicit multi-user signal detection, which increases complexity, introduces latency, and degrades inference performance under adverse channel conditions. Our preliminary experiments reveal that signal detection is not indispensable for multimodal task-oriented SemCom, owing to the inherent dependence and complementarity of semantic information. Motivated by this observation, we propose a multimodal semantic communication framework called DIRECT to achieve multi-user information fusion during transmission, thereby eliminating the need for explicit detection at the receiver. To further enhance task performance and mitigate disturbances from redundant features with unequal contributions, we develop a semantic-weighted importance feature transmission (SWIFT) mechanism. By leveraging SHAP-based analysis, SWIFT selectively transmits high-value features most relevant to the task, effectively reducing semantic redundancy and improving inference accuracy. Experimental results demonstrate that DIRECT enhances the accuracy of video sentiment analysis by about 2%-5% compared with conventional SemCom without signal detection and by about 10% compared with the signal detection scheme.

Keywords:

Semantic communication, task-oriented, multi-user communication, multimodal data, signal detection, Shapley value

Contents

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Denotation

$F_e(\cdot)$	Semantic encoder
$F_d(\cdot)$	Semantic decoder
$F_s(\cdot)$	Feature selection module (FSM)
$C(\cdot)$	Mapping function from real-valued vector to complex-valued signal
i	Index of the uplink users
θ_i^e	Semantic encoder parameters for user i
θ^d	Semantic decoder parameters
ρ_i	Feature selection ratio of FSM for user i
P_i	Power constraint of user i
$\mathbf{x}_i^m$	Input data for user i
$\mathbf{f}_i$	Latent feature produced by the semantic encoder of user i: $F_e(\cdot;\theta_i^e)$
$\mathbf{s}_i$	Transformed latent feature produced by the FSM of user i: $F_s(\cdot;\rho_i)$
$\mathbf{z}_i$	Semantic signal of user i produced by the mapping function C
$\bar{\mathbf{z}}_i$	Normalized semantic signal that satisfies the power constraint
$\mathbf{y}$	Received signal at the base station

r	Final task result
h_i	Fading coefficient at the base station
$\mathbf{n}$	Additive white Gaussian noise at the base station
γ	Semantic signal-to-noise ratio
σ_n^2	Noise power

Chapter 1
Introduction

Conventional wireless communication systems primarily emphasize the accurate and efficient transmission of data over noisy channels. Recently, with the rapid advancements in machine learning and the explosive growth of connected devices, modern networks are generating unprecedented volumes of multimodal data to support a wide range of intelligent applications, such as autonomous driving, real-time translation, and immersive extended reality (XR) services [1], [2]. However, traditional systems have nearly approached Shannon's capacity limit, and finite spectral resources are increasingly insufficient to accommodate the growing data traffic, which is expected to reach 5000 EB per month in 2030 [3]. To address this challenge, semantic communication has emerged as a promising paradigm for enhancing communication efficiency.

Unlike traditional communications that aim to transmit all data exactly, semantic communication (SemCom) focuses on transmitting only the relevant semantic information for the tasks. Leveraging the deep neural networks (DNNs), task-related features can be extracted from the source data and delivered to the receiver. In semantic communication, the emphasis shifts from preserving exact data representations to accurately interpreting the underlying meaning, thereby discarding irrelevant information and enhancing communication efficiency significantly.

In the literature, semantic communication has been categorized into single modal multi-user semantic communication and multimodal multi-user semantic communication. In the single modal setting, multiple users transmit data of the same modality and cooperatively execute an intelligent task [4], [5], [6]. In contrast, multimodal multi-user semantic communication assumes that each user provides data from a single modality. The receiver then fuses these heterogeneous modalities, which complement one another, to accomplish the target task [7], [8], [9], [10], [11], [12].

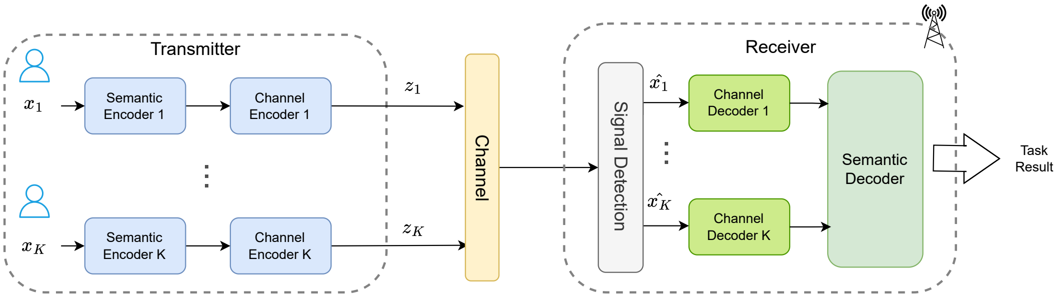


Figure 1.1: Typical Multi-user Semantic Communication System.

In conventional multi-user communication, simultaneous transmissions necessitate multi-user signal detection (MUSD) at the receiver, as depicted in Figure 1.1. After recovering the individual signals, the receiver may either complete the tasks of individual users or fuse the recovered data to address a joint task, with the flexibility to focus on specific portions of the received information. However, such detection procedures often incur significant drawbacks in the context of task-oriented semantic communication. Unlike conventional systems, which emphasize the faithful reconstruction of transmitted features or raw input data, task-oriented semantic communication seeks to directly accomplish inference tasks by transmitting only task-relevant information. This task-specific representation often lacks the redundancy necessary for accurate feature or signal recovery, resulting in imperfect separation and degraded task performance. Furthermore, as the number of users grows, the computational complexity and end-to-end latency associated with multi-user signal detection become increasingly prohibitive [13]. These challenges motivate a fundamental question: Is explicit signal detection necessary in task-oriented semantic communication systems?

An alternative perspective is that explicit signal detection may not always be necessary. In multi-user task-oriented semantic communication systems, the objective is to obtain the final task result rather than to recover transmitted features or reconstruct the raw input data. Furthermore, when multiple users collaborate on a common task, the semantic information they transmit is inherently correlated and complementary, making the separation of individual signals redundant. Consequently, traditional signal detection is not essential for the considered systems.

Although several studies have explored bypassing signal detection in multi-user semantic communication [6], [14], [15], most of them are limited to homogeneous modality scenarios, while research on truly multimodal data remains underexplored. Furthermore, in multimodal systems, semantic features do not contribute equally to task performance, and redundant or invariant features may interfere with the decision-making of the decoder. In the absence of signal detection, however, the receiver can only passively decode the already superimposed information from the channel, lacking the ability to selectively determine which feature to use by attention or distortion mechanisms across modalities. To overcome these challenges, we propose a new framework in which multimodal features are selectively filtered before being transmitted and superimposed in the channel. This strategy ensures that only task-relevant and high-value features are delivered, thereby enhancing both efficiency and robustness in multi-user multimodal semantic communication systems. In summary, while prior studies have made valuable contributions, existing approaches are not well-suited for multimodal semantic communication without signal detection. The main contributions of our work can be summarized as follows:

- To the best of our knowledge, we are one of the first to propose a multi-user multimodal semantic communication framework, i.e., detection-free importance-aware representation for efficient cross-modal transmission (DIRECT), that enables the receiver to accomplish intelligent tasks using the fused semantic information directly, without the need for explicit multi-user signal detection.

- To maximize the task performance, we propose a semantic importance-aware multimodal transmission (SIMT) mechanism to improve the system robustness. A SHAP-based evaluation method is designed to select features that are most important to the corresponding task.

- To empirically evaluate the effectiveness and efficiency of DIRECT, experiments are conducted under two multimodal datasets. The results demonstrate that compared to baseline schemes, DIRECT enhances both robustness and task performance across different channel conditions. In addition, ablation studies are conducted to further validate the core design of our selection method.

Chapter 2

Related Works

In this section, we first provide an overview of semantic communication, covering its core concepts, categories, and multiple-user schemes. Lastly, we discuss the necessity of multi-user signal detection (MUSD) in multimodal semantic communication, where multiple users transmit multimodal data and cooperatively complete an intelligent task.

2.1 Semantic Communication

Unlike traditional communication systems that aim to transmit all bits of data faithfully, semantic communication (SemCom) focuses on delivering only the information relevant to the conveyed meaning. Consequently, the design objective shifts from minimizing bit errors to minimizing semantic errors. To this end, joint semantic–channel coding (JSCC) schemes have been proposed [16], [17], laying the foundation for semantic communication. JSCC systems have been shown to exhibit higher robustness against channel noise, effectively alleviating the cliff effect that arises in conventional systems at low signal-to-noise ratios (SNRs).

Existing research on SemCom can be broadly categorized into *semantic-oriented* and *task-oriented* communication. In semantic-oriented communication [16], [18], [19], [20], [21], [22], [23], [24], the goal is to transmit the overall semantics of the source data such that the receiver can reconstruct the original content. Various architectures have been developed for different modalities, including images [16], [18], text [19], [20], speech [21], [22], and video [23], [24].

In contrast, task-oriented communication [25], [26], [27] focuses on extracting and transmitting only task-relevant semantics, which are directly utilized for downstream decision making at the receiver. For example, [25] proposed a JSCC-based framework for image retrieval over noisy channels, supporting edge-server applications such as image query. In [26], a DL-based semantic communication framework was developed for speech recognition, transmitting recognition-related features and recovering text at the receiver. Similarly, [27] introduced a deep learning-based compressive video streaming scheme tailored to real-time semantic segmentation, with the ability to adapt dynamically to time-varying bandwidth conditions.

2.2 Multi-User Multimodal Semantic Communication

Multimodal semantic communication extends the concept of semantic communication by integrating multiple data modalities—such as text, images, and audio—to enhance communication efficiency and reliability [7], [8]. Such integration enriches the conveyed information and strengthens the system's ability to handle complex tasks, with applications across artificial intelligence, autonomous driving, and industrial systems [9], [10], [12].

Despite these advances, studies rely on multi-user signal detection (MUSD) at the receiver. This process requires separating and recovering the signal from each user individually. Under poor channel conditions, however, multi-user signal detection may be inaccurate, leading to performance degradation in the downstream task. Moreover, as the number of users increases, the computational complexity of detection algorithms significantly raises end-to-end latency.

To address these issues, recent works [6], [14], [15] have proposed multi-user semantic communication frameworks that bypass explicit signal detection by exploiting the natural superposition property of wireless channels for data fusion. For instance, [6] introduced maximal coding rate reduction (MCR^2) as a surrogate objective to represent inference accuracy, replacing the conventional detect-then-classify paradigm. In [14], multimodal data from multiple users are fused during transmission and weighted by channel gains. Similarly, [15] presented SC-DAC, a deep learning-based over-the-air computation (AirComp) framework that integrates communication and computation, enabling simultaneous transmission of semantic features from multiple users over a shared channel.

However, these approaches remain limited to homogeneous-modality scenarios, while the study of truly multimodal multi-user semantic communication remains underexplored. Real-world applications often involve multimodal data, where complementary modalities provide richer and more comprehensive semantic representations. Furthermore, although they evaluate communication effectiveness in terms of task performance, the transmitted semantics are typically obtained by compressing features through a fixed neural network, without explicit task-specific optimization. A more promising direction is to selectively filter modality-specific features, discarding irrelevant semantics so that only task-relevant information is fused, thereby enhancing robustness and improving task performance in multi-user multimodal semantic communication systems.

Chapter 3

Preliminary Experiments

As discussed in the previous chapters, when multiple users cooperate to accomplish a common task, the transmitted semantic information is inherently correlated and complementary. This motivates the hypothesis that explicit multi-user signal detection may not be essential for such task-oriented semantic communication scenarios. In this chapter, we conduct a set of preliminary experiments to validate this claim.

3.1 Experiment Setting

We reproduce the U-DeepSC system [12], which transmits multimodal data under an orthogonal multiple access (OMA) scheme. We then extend it to a non-orthogonal multiple access (NOMA) scheme, with and without signal detection, for performance evaluation. Specifically, semantic successive interference cancellation (SIC) [28] is adopted as the signal detection method. The following experiments are conducted:

- Compare the performance between receivers with and without signal detection.
- Remove one of the modalities to explore the influence of the modality ablation.

We consider video sentiment analysis as the target task. The training configuration follows [12], where all semantic models are trained over an additive white Gaussian noise (AWGN) channel with a fixed SNR of 12 dB.

3.2 Results and Observations

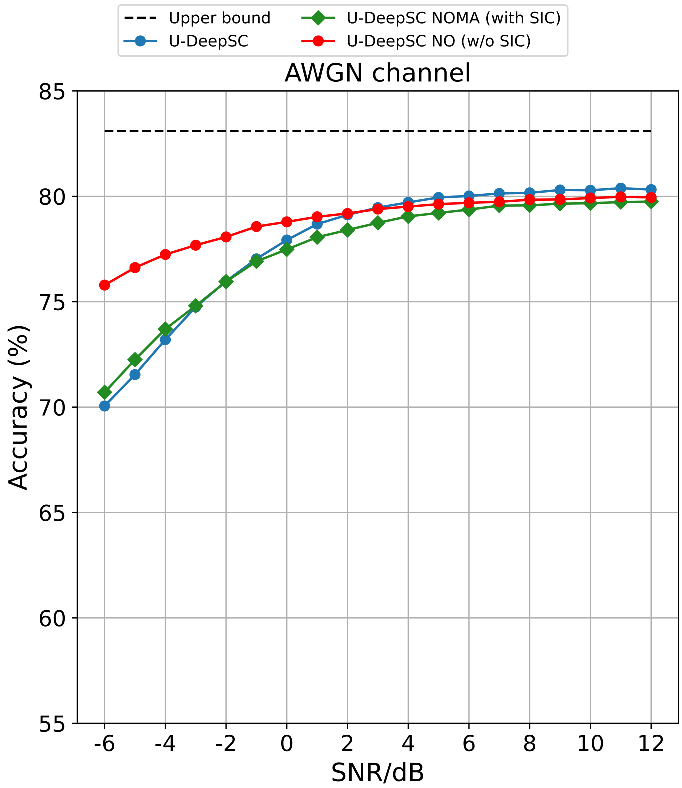
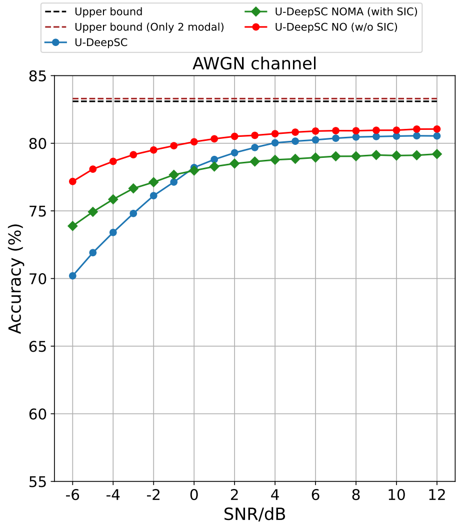


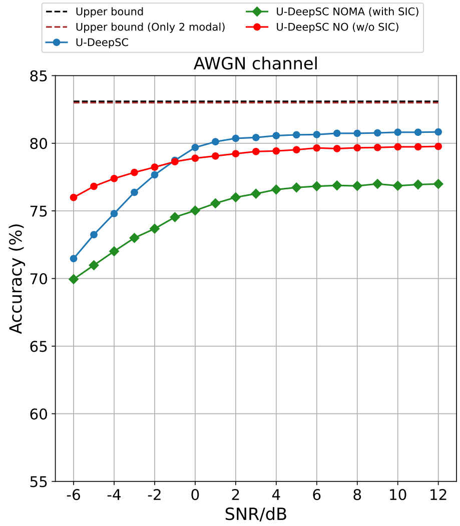
Figure 3.1: Task accuracy comparison on video sentiment analysis

First, as shown in Figure 3.1, the U-DeepSC system without SIC (red curve) consistently outperforms the counterpart with SIC (green curve) under the non-orthogonal scheme, particularly in the low-SNR regime. This result indicates that explicit signal detection fails to accurately separate superimposed signals under poor channel conditions. In contrast, in the absence of explicit signal detection, the unified semantic decoder directly optimizes toward the task objective when decoding the superimposed signal, thereby achieving superior task performance. This observation demonstrates that explicit signal detection is not essential for multi-user task-oriented semantic communication systems.

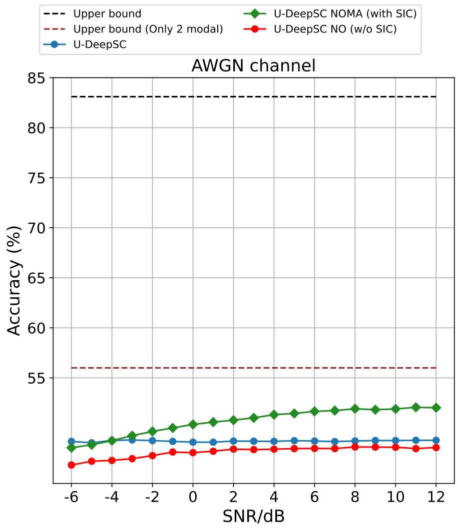
To further evaluate the semantic importance of different modalities in a multi-user superimposed channel, a modality ablation study is conducted, as shown in Figure 3.2. The baseline performance corresponds to the transmission of all three modalities (Text,



(a) Transmit text and image only



(b) Transmit text and speech only



(c) Transmit image and speech only

Figure 3.2: The performance of different schemes when transmitting only two of three modality data for the task

Image, and Speech), as presented in Figure 3.1.

As illustrated in Figure 3.2 (c), the most severe performance degradation occurs when the textual modality is removed. This result empirically demonstrates that text serves as the dominant semantic carrier for the considered task, encoding the most critical information required for the decoder to construct an accurate decision boundary.

Notably, a counterintuitive phenomenon is observed in Figure 3.2 (a) when the speech modality is excluded. Instead of performance degradation, a slight improvement in accuracy is achieved compared with the full-modality baseline. This suggests that, for this specific task, speech signals contribute marginally to the overall semantic representation and may even introduce redundancy.

These observations motivate the core objective of this work: to suppress semantic noise induced by redundant multimodal features in signal-detection-free task-oriented SemCom systems, thereby maximizing end-task performance.

Chapter 4

System Model

4.1 Framework of DIRECT

We consider a multi-user task-oriented semantic communication system, where a series of end-user devices $\mathcal{K} = \{1, 2, \cdots, K\}$ collaborate with a base station (BS) to accomplish an inference task. Inspired by semantic-aware communication paradigms, we propose a feature selection-based multimodal semantic communication framework, i.e., DIRECT, as illustrated in Figure 4.1. Each user is equipped with a semantic encoder, which is jointly trained at the BS. The encoder extracts task-related semantic features from raw data and selectively transmits them through an importance-aware feature selection module (FSM). At the receiver, a single semantic decoder processes the aggregated multi-user features received over the channel and obtain the fused information for the downstream task.

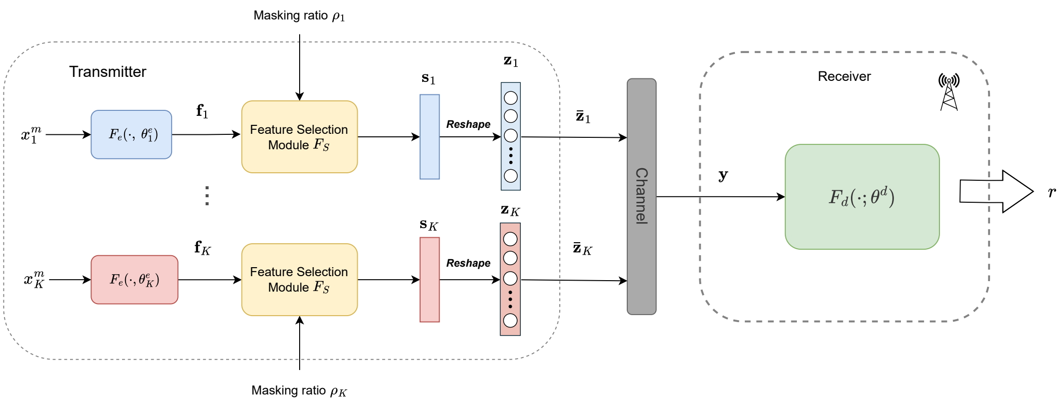


Figure 4.1: Framework of feature selection-based multimodal semantic communication system.

4.1.1 Semantic Transmitter

Let $X = \{x_1^m, x_2^m, \ldots, x_K^m\}$ denote the data collected by edge users, where c is the modal type of data. Each user is a single modal data source. Subsequently, these data are processed to obtain the corresponding encoded features $\mathbf{f} = \{\mathbf{f}_1, \ldots, \mathbf{f}_K\}$. Denote by $\mathbf{f}_i \in \mathbb{R}^{2L}$ the semantic features of user i with dimension L. These semantic feature is first extracted by

$$\mathbf{f}_i = F_e(x_i^m; \theta_i^e), \tag{4.1}$$

where $F_e(\cdot; \theta_i^e)$ represent the semantic encoder with the trainable parameter θ_i^e. Since the receiver will directly perform semantic decoding on the fused semantic features to obtain task results in our system, the existence of redundant or invariant features may mislead the semantic decoder. Thus, we design an importance-aware feature selection module (FSM) which can prioritize the semantically-matched features based on the importance value of features.

Given the selection ratios $\rho = \{\rho_1, \ldots, \rho_i, \ldots, \rho_K\}$ and the semantic importance of each feature from the BS (see Section 5), the feature selection vectors $M = \{\mathbf{m_1}, \ldots, \mathbf{m_i}, \ldots, \mathbf{m_K}\}$ can locate the features of user i to be transmitted according to their importance order (value in $\mathbf{m_i}$ is 1 means the feature need to be transmitted, 0 otherwise). The importance evaluation method will be detailed in the next section. The masked semantic features $\mathbf{S} = \{\mathbf{s}_1, \ldots, \mathbf{s}_K\}$, $\mathbf{s}_i \in \mathbb{R}^{2L}$ to be processed can be represented as:

$$\begin{aligned} \mathbf{s}_i &= F_S(\mathbf{f}_i; \rho_i) \\ &= \mathbf{f}_i \odot \mathbf{m_i}, \end{aligned} \tag{4.2}$$

where $F_S(\cdot, \rho)$ is the FSM and $\odot$ is the Hadamard product, which multiplies the corresponding elements of two matrices or vectors element-wise.

The masked features **S** are then reshaped to the complex-valued semantic signals $\mathbf{z} = \{\mathbf{z}_1, \ldots, \mathbf{z}_K\}$, $z_i \in \mathbb{C}^L$, which is transmitted over the wireless channel and can be represented as:

$$\{\mathbf{z}_1, \ldots, \mathbf{z}_K\} = \{C_1(\mathbf{s}_1), \ldots, C_K(\mathbf{s}_K)\}. \tag{4.3}$$

where $C(\cdot) : \mathbb{R}^{2L} \to \mathbb{C}^L$ is the mapping function.

4.1.2 Wireless Channel

The reshaped semantic signals are sent to the wireless channel after power normalization. We assume both the base station and users use single-antenna setup. The received signal at the base station is modeled as:

$$\mathbf{y} = \left(\sum_{i=1}^{n} \sqrt{P_i} \mathbf{z}_i \right) + \mathbf{n}, \tag{4.4}$$

where P_i is the transmit power of the i_{th} user and $\mathbf{n} \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise that the base station receives; SNR γ is $\sum\limits_i \|\mathbf{z}_i\|^2 / \sigma^2$.

4.1.3 Semantic Reciever

At the receiver, signal detection and separation are not required, the semantic features are fused in wireless channels. The fused semantic feature is then used directly for the inference task:

$$r = F_d(\mathbf{y}, \theta^d) \tag{4.5}$$

where $F_d(\cdot; \theta^d)$ is the decoder parameterized by θ^d, and r denotes the task result. The downstream task may involve semantic segmentation, classification, object detection, or other inference objectives, depending on the application.

Chapter 5

Problem Statement

Since semantic features extracted from raw data exhibit unequal levels of importance, indiscriminately transmitting all features may be ineffective for task inference and lead to interference with other information in the absence of signal detection. Moreover, transmissions that merely take semantic relevance into account may result in unreliable communication links. Thus, a semantic-weight importance-feature transmission (SWIFT) mechanism is proposed to determine the feature transmission ratio and perform feature selection by jointly considering both semantic importance and channel conditions.

5.1 Semantic-Weighted Importance Feature Transmission

To illustrate the operation of the proposed system, we outline the workflow as follows. Note that semantic encoders and the decoder are jointly pre-trained at the BS, where feature importances are also evaluated before deployment. Considering the i-th user, the procedure is: user extracts semantic features $\mathbf{f}_i$ from data x_i^m. The BS determines the transmission ratio decision vector $\rho = \{\rho_1, \ldots, \rho_K\}$ based on channel conditions and historical data distribution. The transmission ratio decision vector ρ_i is then fed back to the FSM of user i to determine the features to be transmitted according to their importance order. Next, The selected feature vectors $\mathbf{s}_i$ are reshaped and normalized into $\bar{\mathbf{z}}_i$. The normalized semantic signals $\bar{\mathbf{z}}_i$ are transmitted and naturally superimposed in the wireless channel. Finally, At the receiver, explicit signal detection and separation are no longer required; instead, the fused semantic features $\mathbf{y}$ are directly decoded by the semantic decoder to obtain task results r.

This mechanism ensures that only task-relevant and channel-robust features are transmitted, thereby improving both efficiency and reliability in multi-user multimodal semantic communication.

Finally, it is essential to note that in task-oriented semantic communication, conventional link-level metrics, such as bit error rate or throughput, fail to capture the ultimate objective. Instead, semantic-level performance indicators, such as task accuracy, are adopted.

We denote by η the semantic task accuracy, which depends on both the feature selection ratios $\rho = (\rho_1, \cdots, \rho_K)$ across K modalities and the channel condition represented by the signal-to-noise ratio (SNR) γ. Thus, η can be expressed as a multivariate function $\eta = g(\rho_1, \cdots, \rho_K, \gamma)$

To approximate $g(\cdot)$, inspire by the ideas in [29], we adopt polynomial regression due to its interpretability and simplicity. Polynomial regression also provides a flexible yet tractable way to capture nonlinear dependence of task accuracy on both feature selection ratios and SNR. Here, we take quadratic polynomial approximation as an example.

In general, a polynomial regression of degree d can approximate the semantic task accuracy function as:

$$\eta \approx g(\rho_1, \ldots, \rho_K, \gamma) = \sum_{|\alpha| \leq d} \beta_\alpha \, \rho_1^{\alpha_1} \cdots \rho_K^{\alpha_K} \gamma^{\alpha_{K+1}} + \epsilon, \tag{5.1}$$

where $\alpha = (\alpha_1, \ldots, \alpha_{K+1})$ is a multi-index with non-negative integers and $|\alpha| = \sum_{i=1}^{K+1} \alpha_i \leq d$. The coefficients β_α are estimated from experimental data.

The regression problem can be expressed in matrix form as:

$$\mathbf{y} \approx \mathbf{V}\boldsymbol{\beta},$$

where $\mathbf{y} \in \mathbb{R}^N$ is the vector of observed accuracies from N experiments, $\mathbf{V} \in \mathbb{R}^{N\times T}$ is the design matrix containing all polynomial features up to degree d, $\boldsymbol{\beta} \in \mathbb{R}^T$ is the coefficient vector, and T is the total number of polynomial terms. The least-squares estimate is

$$\boldsymbol{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.$$

In practice, we obtain $\mathbf{y}$ by conducting model inference on validation dataset under different SNRs and feature selection ratios. Note that, training the semantic communication model is executed in the BS, thus calculating such function using a numerical approach is possible. By solving the regression problem, we obtain an explicit approximation of $g(\rho_1, \cdots, \rho_K, \gamma)$, which allows us to predict and optimize semantic task accuracy under various conditions.

5.2 Problem Formulation

We aim to optimize the feature selection ratios across K users to maximize the semantic task performance under the considered system architecture. The optimization problem is formulated as

$$\begin{aligned}\max_{\rho_1,\cdots,\rho_K} \quad & \eta \\ \text{s.t.} \quad & 0 \leq \rho_i \leq 1, \quad \forall i \in \mathcal{K}, \\ & P_i \leq P_i^{\max}, \quad \forall i \in \mathcal{K},\end{aligned}$$

where $P_i^{\max}$ is the power constraint of the i_{th} user.

We aim to optimize this problem under the following constraint:

- The receiver do not separate the received signal to each user signal, so it only have the global semantic information of all users.
- In practice, the users do not know other user's information during communication, each user only coordinate with the receiver. So the attention method that capture important correlation across multimodal data can not apply here.

Chapter 6
Methodology

6.1 SHAP-Based Feature Selection Method

To quantify the semantic importance of individual features for the downstream task, we employ a SHAP-based evaluation metric.

SHAP values [30] provide a model-agnostic interpretability framework, in which each prediction is explained by the contribution of input features to the model output. Specifically, SHAP values approximate Shapley values [31], a concept from cooperative game theory that ensures a fair distribution of rewards according to each player's marginal contribution. In machine learning, Shapley values measure the impact of each feature on overall task performance. A model f_M is viewed as a coalition game with M features or modalities as players. The Shapley value ϕ of a feature i is calculated as the weighted average of its marginal contributions to all possible coalitions:

$$\phi(M, f_M) = \sum_{S \subseteq M \setminus \{i\}} \frac{|S|!(|M| - |S| - 1)!}{|M|!}(v(S \cup \{i\}) - v(S)) \tag{6.1}$$

Since exact computation is combinatorial and computationally prohibitive, SHAP employs efficient approximations, such as weighted linear regression for general models or distributional assumptions for tree-based models [30]. We reasoned that SHAP highlights the contribution of each feature to model decisions, we leverage it as the basis for feature selection. In the proposed SWIFT mechanism, the task importance of encoded features **f** is quantified using SHAP values, and the most important features are prioritized for transmission.

6.2 Optimization

Once we obtain the approximation of $g(\rho_1, \cdots, \rho_K, \gamma)$ through polynomial regression, the feature selection ratios can be optimized to maximize semantic task accuracy under a given channel condition. Specifically, for a fixed SNR γ, the regression function reduces to

$$\eta(\rho) \approx g(\rho_1, \ldots, \rho_K \mid \gamma), \tag{6.2}$$

which becomes a nonlinear function of the feature selection ratios $\rho = (\rho_1, \ldots, \rho_K)$ only.

The optimization problem is then expressed as

$$\begin{aligned} \max_{\rho} \quad & g(\rho_1, \ldots, \rho_K \mid \gamma) \\ \text{s.t.} \quad & 0 \leq \rho_i \leq 1, \quad \forall i \in \mathcal{K}, \\ & P_i \leq P_i^{\max}, \quad \forall i \in \mathcal{K}, \end{aligned} \tag{6.3}$$

Since the objective function is smooth and the constraints are continuous, Sequential Least Squares Quadratic Programming (SLSQP) is adopted to solve Equation (6.3). SLSQP is particularly suitable because it can handle nonlinear objectives with both equality and inequality constraints, and guarantees convergence to a local optimum under mild regularity conditions. The optimization can be efficiently executed at the base station after model training, since the regression function provides an explicit surrogate model of the task performance. It is worth mentioning that, once the semantic communication network is trained, the regression coefficients β_α remain fixed, and thus the optimization of feature selection ratios only requires solving Equation (6.3) once for each channel condition, incurring negligible computational overhead during deployment.

Algorithm 1 SLSQP-Based Optimization of Feature Selection Ratios

1: **Inputs:** trained encoders/decoder; ploynominal degree d; SNR γ; feature importance vectors

2: **Outputs:** $\rho^\star \in [0,1]^K$, predicted accuracy $\hat{\eta}^\star$.

3: Define surrogate $\hat{g}(\rho;\gamma) = \sum_{|\alpha|\le d} \beta_\alpha \rho_1^{\alpha_1} \cdots \rho_K^{\alpha_K} \gamma^{\alpha_{K+1}}$ with γ fixed.

4: **Objective:** $f(\rho) = -\hat{g}(\rho;\gamma)$ (convert maximization to minimization).

5: **Constraints:***Bounds:* $0 \le \rho_i \le 1$.

6: **Initialization:** $\rho^{(0)} \leftarrow$ normalized SHAP-based importance (or $\rho^{(0)} = \mathbf{1}$ / uniform).

7: **SLSQP loop:** Starting from $\rho^{(0)}$, iteratively:

   1. Form a quadratic subproblem using local linearization of constraints and (optionally) the analytic gradient $\nabla \hat{g}(\rho;\gamma)$.
   2. Solve the QP to get a step $\Delta\rho$, apply line search, and project onto bounds.
   3. Update Lagrange multipliers; check KKT residual and constraint violations.

8: Stop when $\|\nabla f(\rho) + \sum_i \lambda_i \nabla c_i(\rho)\|_\infty \le \varepsilon$ and all constraints satisfied within tolerance.

9: **Return** $\rho^\star$, $\hat{\eta}^\star = \hat{g}(\rho^\star;\gamma)$.

Chapter 7

Experiments

In this chapter, we evaluate the performance of the DIRECT framework the and SWIFT mechanism. We consider a scenario where the base station (BS) jointly pre-trains the semantic encoders and decoder. The pre-trained models are subsequently distributed to the users, who then collaborate with the BS to optimize the feature selection module (FSM) under the current channel condition. We first describe the experimental setup, including datasets, models, and communication scenarios. The proposed scheme is then compared with baseline approaches that bypass signal detection but do not employ importance-aware feature selection.

7.1 Experimental Setup

In the experiments, we take the multimodal sentiment analysis and audio visual event tasks as examples. We conduct experiments on CMU-MOSEI [32] and AVE dataset [33] for multimodal sentiment analysis and audio visual event tasks, respectively. CMU-MOSEI, consisting of text, image, and speech modalities, is the largest dataset of multimodal sentiment analysis and emotion recognition to date. It contains more than 23,500 sentence utterance videos. All the sentence utterances are randomly chosen from various topics and monologue videos. The AVE dataset contains 4143 videos covering 28 event categories, and videos in AVE are temporally labeled with audio-visual event boundaries.

For the text data, we utilize BERT-base-uncased as the text embedding tool. For the video data's vision stream, Facet, an analytical tool based on the facial action coding system (FACS), is used to extract facial features. Speech features are extracted using COVAREP, a professional acoustic analysis framework. Furthermore, throughout the simulations, we assume an AWGN channel.

We compare DIRECT against the following schemes:

- **Error-free transmission(labeled as "Upper bound"):** Full noiseless data is delivered to the receiver, serving as the performance upper bound.
- **OMA SemCom method (labeled as "OMA"):** Users transmit semantic information through orthogonal channels, e.g., TDMA, FDMA. All settings follow the work [12].
- **Non-orthogonal SemCom with signal detection (labeled as "w/ Signal Detection"):** Users share the same physical resources, and the receiver performs semantic signal detection [28] prior to feature fusion.
- **Non-orthogonal SemCom without signal detection (labeled as "w/o Signal Detection"):** Users share the same physical resources, and the receiver performs information fusion using the fused semantic information directly.

7.1.1 Semantic Models

Specifically, we use the Bert model as the text encoder and Vision Transformer as the image and speech encoders. The semantic decoder follows the architecture proposed in [12]. All models are jointly trained for 150 epochs. The settings of the training are the AdamW optimizer with learning rate $3{\times}10^{5}$ initially, and drop to 1×10^{-5} after 10 epochs, batch size 50, and weight decay $5{\times}10^{-31}$. The image and speech semantic encoders are initialized using a pretrained vision transformer, while the text semantic encoder is initialized by the pre-trained BERT model.

All experiments are implemented in PyTorch. Training and evaluations are conducted on an Ubuntu 24.04 system equipped with an Intel Xeon W5-3435X CPU and an NVIDIA RTX 4090 GPU.

7.2 Effectiveness of Proposed Framework

7.2.1 Multimodal Sentiment Analysis (MSA)

Sentiment analysis seeks to identify and interpret sentiment polarity from available information and has been widely applied in domains such as public governance, business intelligence, healthcare, and marketing. To enhance the accuracy and robustness of sentiment inference, multiple users may simultaneously provide complementary observations from diverse viewpoints and heterogeneous modalities. In this setting, the proposed semantic communication framework is employed to transmit task-relevant semantic information, thereby reducing communication overhead while ensuring efficient and reliable information exchange.

We evaluate the task accuracy of the investigated schemes over a wide range of channel conditions, with the signal-to-noise ratio (SNR) varying from -5 to 15 dB under both AWGN and Rayleigh fading channels. The corresponding results are presented in Figures 7.1 and 7.2. All models are trained at an SNR of 12 dB to ensure a fair comparison. In the OMA scheme, each user transmits 16 symbols, whereas the non-orthogonal schemes share the same radio resource and jointly transmit a total of 48 symbols.

Under the AWGN channel, DIRECT consistently achieves the highest task accuracy across the entire SNR range, with particularly pronounced gains in low-SNR regimes.

Compared with the baseline scheme without signal detection, the proposed approach exhibits significantly improved robustness to channel noise. This performance advantage can be primarily attributed to the feature selection module (FSM), which effectively prioritizes task-relevant semantic features while suppressing redundant or less informative components during multimodal fusion. In contrast, the baseline schemes indiscriminately transmit all features, including redundant ones, which may interfere with one another and consequently degrade task performance. Notably, DIRECT even outperforms the OMA scheme at high SNRs, despite the absence of multi-user interference in OMA. This observation suggests that, beyond channel interference, redundant semantic features themselves can act as semantic noise. By retaining only discriminative information, the FSM enables more efficient semantic representation and improves the effectiveness of downstream task inference.

For the Rayleigh fading channel, the overall task accuracy is slightly reduced compared to the AWGN case due to the increased channel uncertainty and fading effects. Nevertheless, DIRECT consistently maintains an accuracy above 70% across all SNRs and outperforms both the schemes with and without signal detection. Moreover, its performance remains competitive with that of the OMA scheme over the entire SNR range, demonstrating strong robustness under more challenging channel conditions.

Overall, these results validate the effectiveness of DIRECT in enhancing both robustness and task performance across diverse channel environments, highlighting the importance of semantic-aware feature selection in multi-user multimodal SemCom systems.

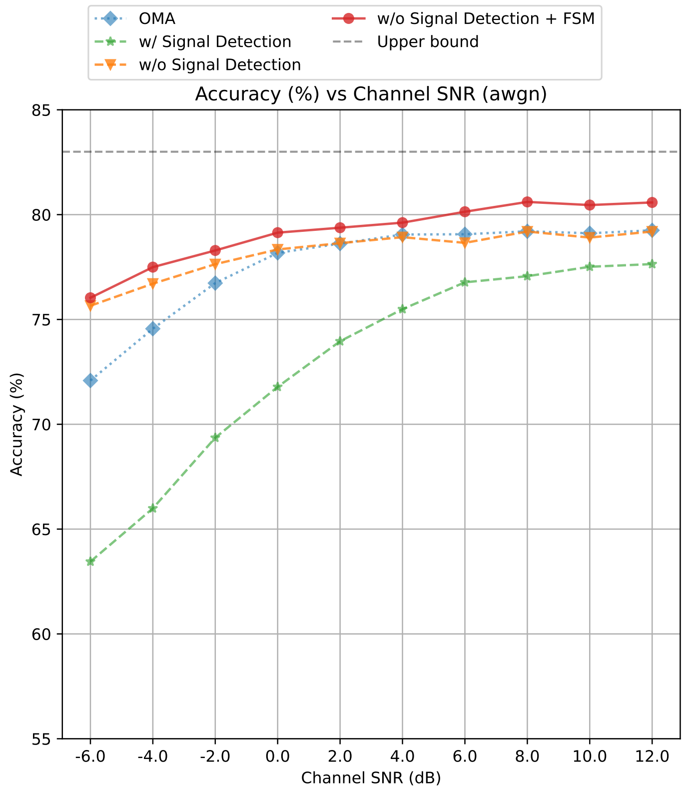


Figure 7.1: Task accuracy on MSA task under AWGN channel.

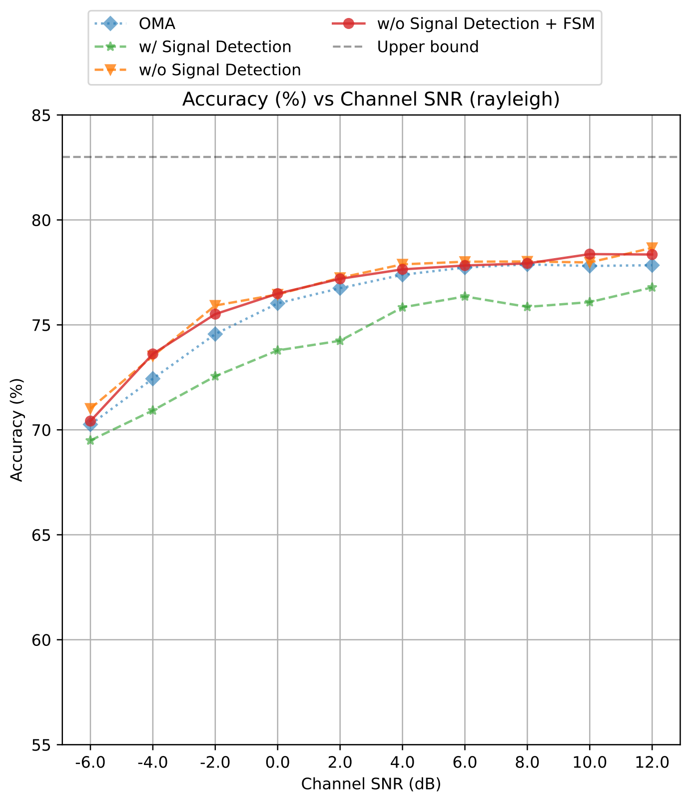


Figure 7.2: Task accuracy on MSA task under Rayleigh fading channel.

7.2.2 Audio Visual Event (AVE)

In this subsection, we apply DIRECT on the audio-visual event (AVE) classification task [33] to evaluate its performance. The considered task involves two modalities, namely visual and audio signals. The wireless channel model, semantic encoders, and decoder remain consistent with those used in the action recognition task. All models are trained for 100 epochs with an initial learning rate of 0.01. Training is conducted at an SNR of 12 dB. In the OMA scheme, each user transmits 24 symbols, while the non-orthogonal schemes transmit 48 symbols in total. During testing, the SNR ranges from -6 dB to 12 dB.

To demonstrate the effectiveness of our framework, we compare it with other schemes under an AWGN channel environment, and the results are presented in Figure 7.3. As shown in Figure 7.3, within the low-SNR range of −6 to 2 dB, DIRECT achieves significantly higher accuracy than both the OMA and the scheme with signal detection, with consistent performance improvements as the SNR increases. Compared to the scheme without signal detection, DIRECT exhibits comparable performance at low SNRs and demonstrates clear advantages at moderate-to-high SNRs. This behavior can be attributed to the increased complexity of the AVE task, which involves classification among 29 event classes. Under severely noisy conditions, certain redundant features may still be beneficial for maintaining classification reliability. As the channel quality improves, however, DIRECT effectively suppresses semantic noise introduced by redundant features, resulting in superior performance.

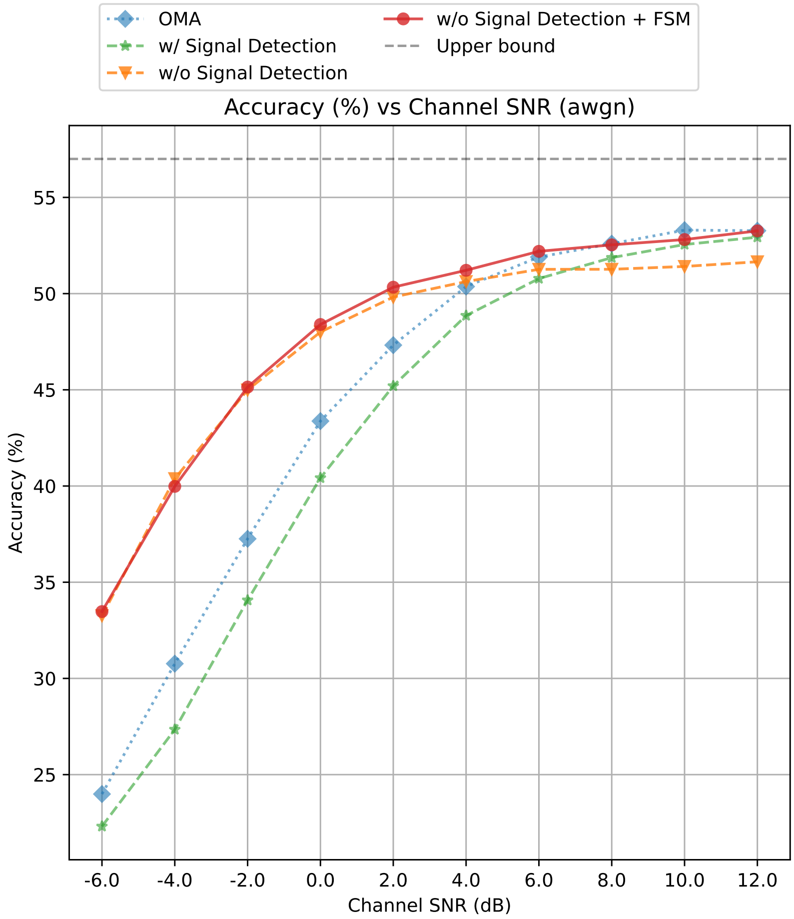
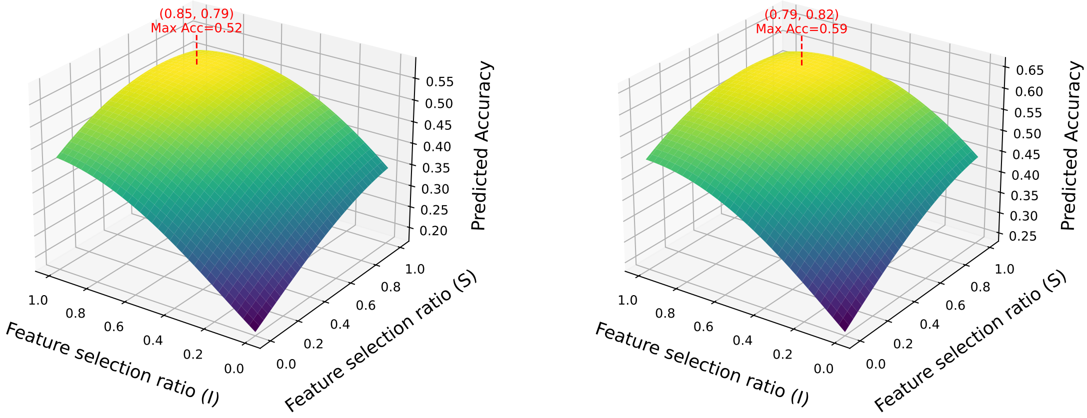


Figure 7.3: Task accuracy on audio video event (AVE) task

7.3 Estimated Model Visualization

By testing the pre-trained SemCom model under different SNR values and feature selection ratios, we can estimate the mathematical relationship between task accuracy, SNR, and selection ratios using the regression method in (5.1), which is shown in Figure 7.4. Notably, peak accuracy is achieved through partial masking rather than full feature transmission. While increasing SNR from 0 dB to 5 dB improves the maximum accuracy from 0.52 to 0.59, the optimal selection ratio remains consistent. This validates our framework's ability to isolate robust semantic features, effectively mitigating channel noise and eliminating the interference of redundant information.



(a) SNR 0 dB (b) SNR 5 dB

Figure 7.4: The estimation model of semantic task accuracy for AVE task under different SNR conditions.

7.4 End-to-End Latency

To illustrate the computational costs of each scheme, we evaluate the end-to-end latency introduced for each communication. Figure 7.5 presents the measured end-to-end latency in a three-user scenario. The baseline scheme with signal detection incurs the highest delay, which can be mainly attributed to the substantial computational complexity required for real-time multi-user signal detection, resulting in a heavy processing burden.

In contrast, the scheme without signal detection achieves a significant latency reduction, with an approximately 20.3% decrease compared to the baseline. DIRECT also exhibits superior efficiency, reducing the end-to-end latency by about 12.4% relative to the baseline. Although the integration of the FSM introduces a modest overhead compared to the scheme without signal detection, it remains considerably more efficient than the full signal detection approach. This efficiency arises because most FSM-related computations, including feature selection and state transitions, are performed during the offline phase, thereby substantially reducing the online processing complexity.

It is worth noting that the results shown in Figure 7.5 correspond to a scenario with three active users. As the computational complexity of signal detection generally increases with the number of users, the latency gap between the baseline scheme and DIRECT is expected to widen further in denser multi-user scenarios.

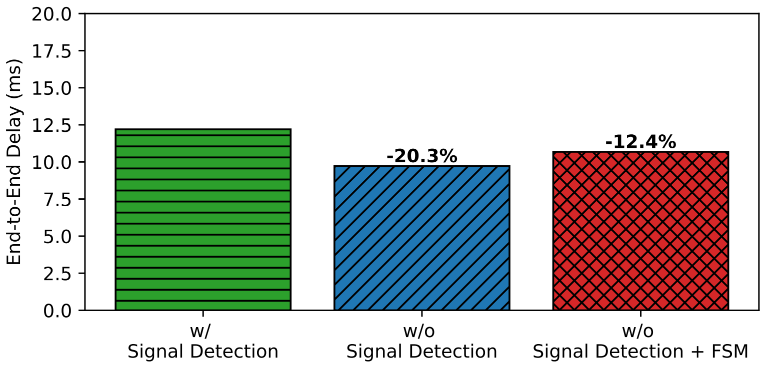


Figure 7.5: Comparison of end-to-end delay across different schemes in a 3-user scenario.

7.5 Ablation Study

To further evaluate the effectiveness of the SHAP-based feature selection method, we conduct an ablation study by comparing it with alternative selection methods. For benchmarking purposes, the estimated model is also trained using data generated by a random feature selection scheme, as illustrated in Figure 7.6. The results demonstrate that the SHAP-based method consistently achieves the highest classification accuracy across all SNR levels and closely approaches the performance upper bound in high-SNR regimes.

In contrast, the random feature selection scheme fails to deliver consistent performance gains and, in some cases, performs even worse than the baseline without signal detection, particularly under low-SNR conditions. This performance degradation arises because randomly discarding features may remove task-critical semantic information, thereby impairing inference accuracy more severely than transmitting all features without selection. These observations highlight the necessity of informed feature selection in semantic communication systems.

Overall, the ablation results confirm the effectiveness of the proposed SHAP-guided framework, which selectively preserves task-relevant features to achieve robust and accurate performance. In comparison, naive random selection undermines the benefits of feature reduction and may even exacerbate semantic information loss.

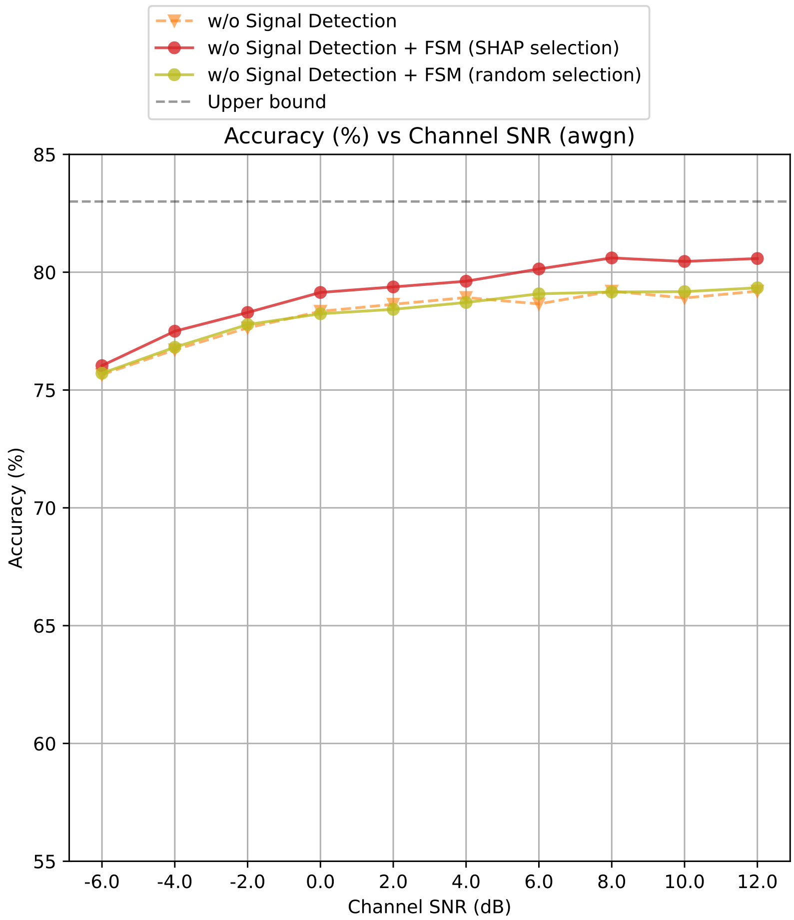


Figure 7.6: Performance of feature selection method

Chapter 8

Discussions

The DIRECT has remarkable performance on task accuracy and end-to-end latency. Besides, there is also some further discussion.

8.1 Improve Communication Efficiency

In DIRECT, redundant features are masked only rather than entirely removed. However, transmitting all features—including masked ones—is impractical for resource-constrained devices. To further enhance communication efficiency, vector quantization represents a promising direction. Specifically, unmasked features can be mapped to discrete tokens using a shared, unified codebook. Nevertheless, under the non-orthogonal scheme without explicit MUSD, the signals from multiple users are superimposed over the channel, and the decoder processes directly on the aggregated signal. As a result, it is crucial for the decoder to correctly infer the positional information of the unmasked features. In our future works, an effective transmission mechanism that preserves essential positional information while reducing the number of transmitted symbols should be explored.

8.2 Resource Allocation

In conventional communication systems, resource allocation is primarily designed to maximize sum capacity or achievable transmission rates by mitigating inter-user interference. In contrast, in the proposed scheme without multi-user signal decoding (MUSD), multiple users transmit their multimodal data over the same radio resources—such as time slots and frequency. Since the transmitted data from different users are complementary rather than competing, such superposition does not introduce harmful interference, which fundamentally distinguishes the DIRECT from traditional resource allocation paradigms.

Consequently, the resource allocation problem in this setting is considerably simplified. Identical radio resources can be assigned to users that collaborate on a common task and whose transmitted data are intended to be jointly fused. Meanwhile, transmissions from different user groups can be allocated orthogonal resources to prevent inter-group interference. This group-based resource allocation strategy effectively balances communication efficiency and system scalability.

Chapter 9

Conclusion

In this thesis, we investigated the necessity of signal detection in task-oriented multi-modal semantic communication systems. A detection-free importance-aware representation for efficient cross-modal transmission (DIRECT) framework was proposed, which exploits the superposition property of signals to fuse heterogeneous information during transmission, thereby eliminating the need for explicit multi-user detection at the receiver. To further enhance task performance, we developed the semantic-weight importance-feature transmission (SWIFT) mechanism, enabling partial feature transmission according to both semantic importance and channel conditions. A SHAP-based feature evaluation method was introduced to quantify feature contributions with theoretical support. Experimental results confirm that the DIRECT consistently outperforms conventional schemes without signal detection across all SNR levels. Moreover, our analysis reveals that even in the absence of multi-user interference, as in the OMA scheme, redundant features still act as semantic noise, underscoring the necessity of feature selection for reliable and efficient semantic communication.

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